



WELCOME



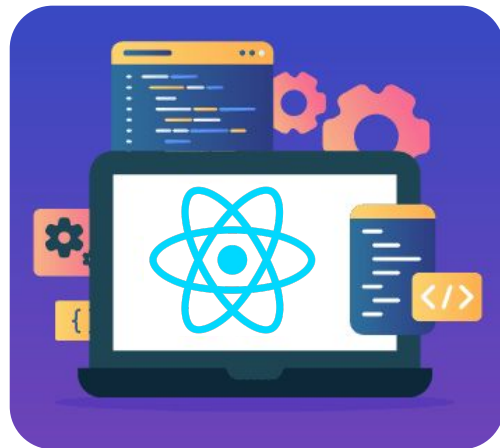


Components and Props

Recap

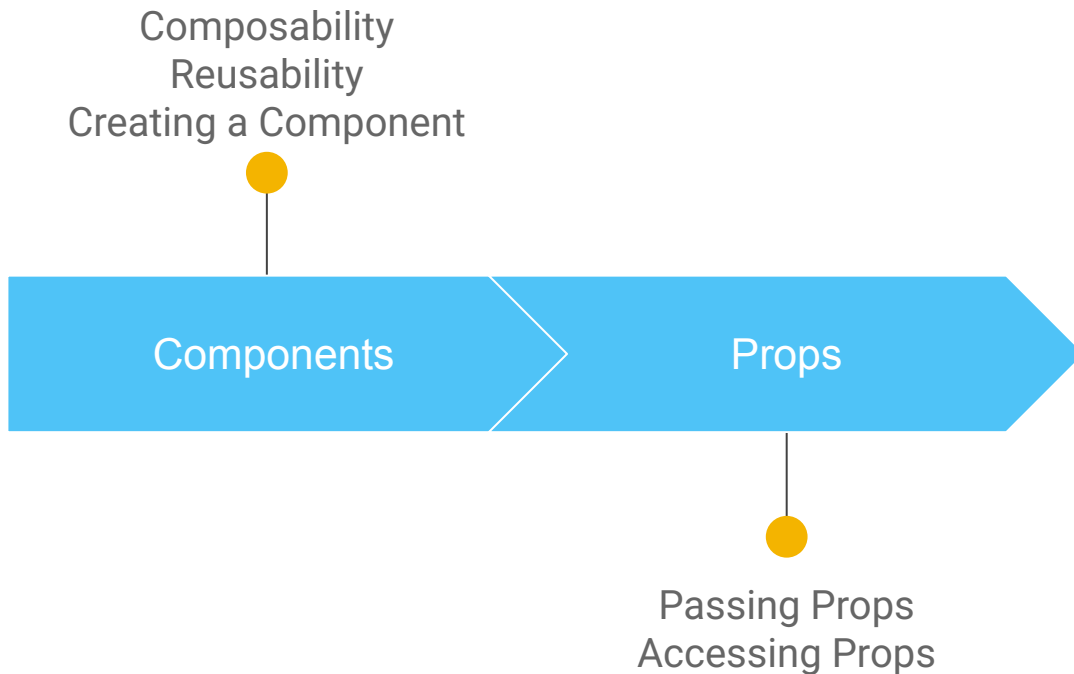
Introduction to React JS

- ✓ MERN Stack
 - Introduction to React JS
- ✓ React
 - `React.createElement()`
 - `ReactDOMClient.createRoot()`
 - `root.render()`
- ✓ JSX
 - Embedding Expressions
 - Nesting elements



Agenda for Today's Session

Components and Props



Introduction to React JS

Session Initial Setup

The screenshot displays the NextWave Learning Platform interface. On the left is a dark sidebar with a list of course items: 'Introduction to React JS | Cheat Sheet' (15 mins), 'Practice 1' (30 mins), 'Solving A React JS Question In Cloud IDE' (2 mins), 'Coding Practice 1' (1 hour), 'Components & Props' (40 mins), 'Components & Props | Cheat Sheet' (15 mins), and 'Practice 2' (30 mins). The main content area has a top navigation bar with a search bar (containing 'Search for \'Arrays\''), a clock icon, and links for 'Home', 'Code Playground', 'Tech Educator', and a user profile. Below the navigation bar is a video player showing a course thumbnail with a play button, a progress bar at 0:10 / 32:49, and controls for volume, language (English), settings, and full screen. Under the video player are links for 'Submit Feedback' and 'Ask doubt'. A tabbed interface below shows 'Notes', 'Code Playground' (selected), 'Discussions', and 'Slides'. The 'Code Playground' tab contains the heading 'Instructions' followed by the text 'Go through the below steps to complete this question:' and a bulleted list: 'Open IDE (It may take 1-2 minutes to initialize the environment)' and 'Practice the concepts and skills you have learned during the session using IDE'. At the bottom center of the page is a blue button labeled 'OPEN IDE' enclosed in an orange rectangular box. A small robot icon is visible in the bottom right corner of the page.

NXT WAVE™

Search for 'Arrays'

Home Code Playground Tech Educator

Introduction to React JS | Cheat Sheet
15 mins

Practice 1
30 mins

Solving A React JS Question In Cloud IDE
2 mins

Coding Practice 1
1 hour

Components & Props
40 mins

Components & Props | Cheat Sheet
15 mins

Practice 2
30 mins

0:10 / 32:49 English

Submit Feedback Ask doubt

Notes Code Playground Discussions Slides

Instructions

Go through the below steps to complete this question:

- Open IDE (It may take 1-2 minutes to initialize the environment)
- Practice the concepts and skills you have learned during the session using IDE

OPEN IDE

Components



React

Component

A Component is a **JS Function** that returns a **JSX Element**

JS

```
const Welcome = () => <h1 className="message">Hello, User</h1>
```

Naming Convention

Component **name** should always start with a **Capital letter**

JS

```
...  
<script type="text/babel" data-type="module">  
  import React from "https://esm.sh/react@19.1.1/?dev";  
  import ReactDOMClient from "https://esm.sh/react-dom@19.1.1/client?dev";  
  
  const Welcome = () => <h1 className="message">Hello, User</h1>  
  const root = ReactDOMClient.createRoot(document.getElementById("root"));  
  root.render(<Welcome />);  
</script>  
...
```


Creating a Component

Output

Hello, User

What if a Component name
starts with LowerCase?

Naming Convention

React treats components starting with **lowercase letters** as **HTML Elements**

JS

```
...  
<script type="text/babel" data-type="module">  
  ...  
  const welcome = () => <h1 className="message">Hello, User</h1>  
  const root = ReactDOMClient.createRoot(document.getElementById("root"));  
  root.render(<welcome/>);  
</script>  
...
```

Component Name with Lowercase

Element Panel Output

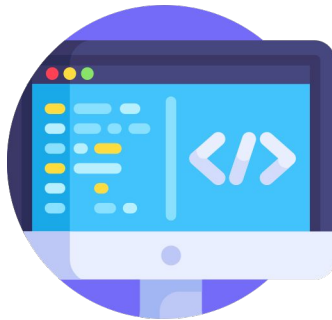
```
<!DOCTYPE html>
<html>
  ▶ <plasmo-csui> ... </plasmo-csui>
  ▶ <head> ... </head>
  ▼ <body>
    ▼ <div id="root">
      ... <welcome></welcome> == $0
    </div>
    ▶ <script type="text/babel" data-type="module"> ... </script>
  </body>
</html>
```

Component Name with Lowercase

Console Output

✖ ▶ Warning: react-dom.development.js:61
The tag <welcome> is unrecognized in this browser. If you meant to render a React component, start its name with an uppercase letter.

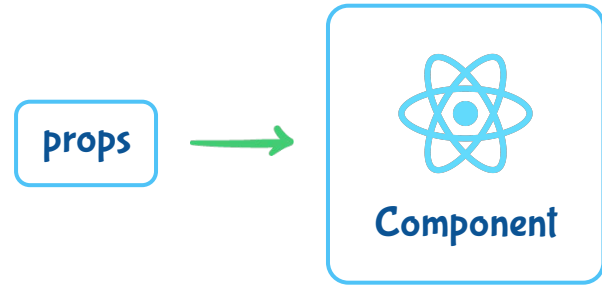
Properties (Props)



React Component

Properties (Props)

React allows us to **pass information** to a Component using **props**



React Component

Passing Props

We can pass **props** to any **Component**
as we declare **attributes** for any **HTML tag**

```
<Component propName1="propValue1" propName2="propValue2" />
```


Passing Props

JS

...

```
const Welcome = () => <h1 className="message">Hello, User</h1>  
const root = ReactDOMClient.createRoot(document.getElementById("root"));  
root.render(<Welcome name="Rahul" greeting="Hello" />);
```

...

Accessing Props

Components accept **props** as **parameters** and can be accessed directly

JS

```
const Component = (props) => {  
  // We can access props here  
}
```

Accessing Props

JS

```
...  
const Welcome =(props) => {  
  console.log(props);  
  return <h1 className="message">Hello, User</h1>;  
};  
const root =  
ReactDOMClient.createRoot(document.getElementById("root"));  
root.render(<Welcome name="Rahul" greeting="Hello" />);  
...
```

```
{name: "Rahul", greeting: "Hello"}  
  i  
  greeting: "Hello"  
  name: "Rahul"  
  ► __proto__: Object
```

Accessing Props

JS

```
...  
const Welcome =( props ) => {  
  const { name, greeting } = props;  
  return <h1 className="message">{greeting}, {name}</h1>;  
};  
const root =  
ReactDOMClient.createRoot(document.getElementById("root"));  
root.render(<Welcome name="Rahul" greeting="Hello" />);  
...
```

Accessing Props

Output

Hello, Rahul

React Component

Reusable

A Component is a piece of **reusable code** that can be used in various parts of an application



Reusable

JS

```
...  
  
const Welcome =( props ) => {  
  const { name, greeting } = props;  
  return <h1 className="message">{greeting}, {name}</h1>;  
};  
  
const root = ReactDOMClient.createRoot(document.getElementById("root"));  
root.render(  
  <div>  
    <Welcome name="Rahul" greeting="Hello" />  
    <Welcome name="Ram" greeting="Hi" />  
  </div>  
);  
...
```

React Component

Output

Hello, Rahul

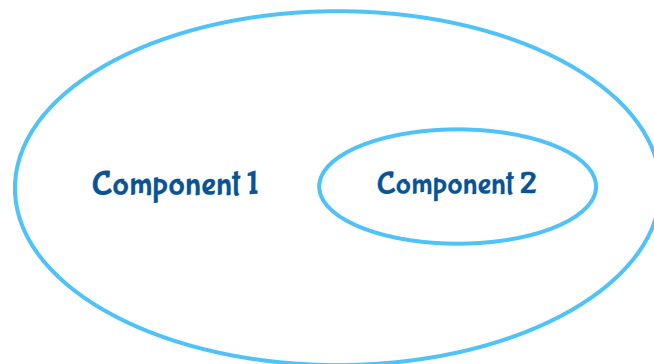
Hi, Ram

React Component

Composable

Components are composable

We can include a **component**
inside another **component**



Composable

JS

```
...  
const Welcome = (props) => {  
  const { name, greeting } = props;  
  return (<h1 className="message">{greeting}, {name}</h1>);  
}  
  
const Greetings = () => (  
  <div>  
    <Welcome name="Rahul" greeting="Hello" />  
    <Welcome name="Ram" greeting="Hi" />  
  </div>  
)  
  
const root = ReactDOMClient.createRoot(document...("root"));  
root.render(<Greetings/>);  
...
```

React Component

Output

Hello, Rahul

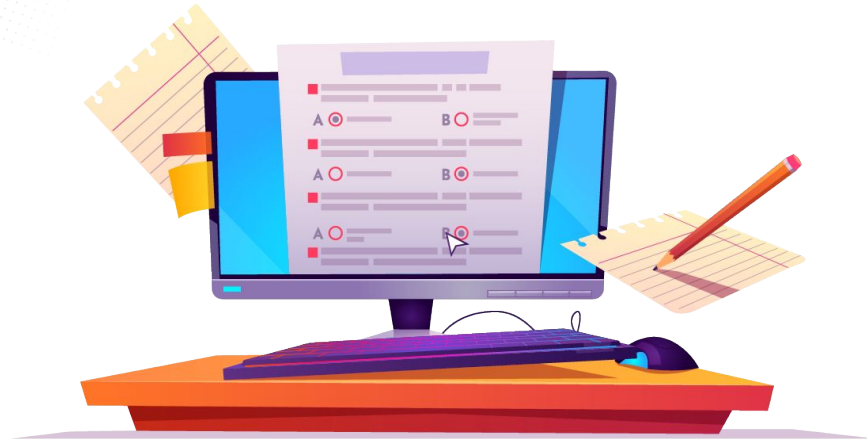
Hi, Ram

Key Takeaways

- ✓ Components
 - Creating a Component
 - Reusable Components
 - Composable Components
- ✓ Props
 - Passing Props
 - Accessing Props



Practice





Next Session

Lists & Keys



THANK YOU





ALL THE BEST

